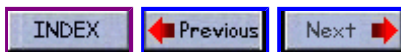
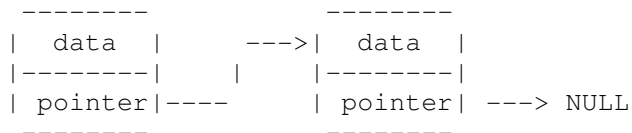


C Programming



LINKED LISTS

A linked list is a complex data structure, especially useful in systems or applications programming. A linked list is comprised of a series of nodes, each node containing a data element, and a [pointer](#) to the next node, eg,



A structure which contains a data element and a pointer to the next node is created by,

```

struct list {
    int    value;
    struct list *next;
};

```

This defines a new data structure called *list* (actually the definition of a node), which contains two members. The first is an integer called *value*. The second is called *next*, which is a pointer to another list structure (or node). Suppose that we declare two structures to be of the same type as *list*, eg,

```

struct list  n1, n2;

```

The next pointer of structure *n1* may be set to point to the *n2* structure by

```

/* assign address of first element in n2 to the pointer next of the n1 st
n1.next = &n2;

```

which creates a link between the two structures.

```

/* LLIST.C    Program to illustrate linked lists */
#include <stdio.h>
struct list {
    int          value;
    struct list *next;
};

main()
{
    struct list n1, n2, n3;
    int         i;

    n1.value = 100;
    n2.value = 200;
    n3.value = 300;

```

```
    n1->next = &n2;
    n2->next = &n3;
    i = n1->next->value;
    printf("%d\n", n2->next->value);
}
```

Not only this, but consider the following

```
n1->next = n2->next      /* deletes n2 */
n2_3->next = n2->next;    /* adds struct n2_3 */
n2->next = &n2_3;
```

In using linked list structures, it is common to assign the value of 0 to the last pointer in the list, to indicate that there are no more nodes in the list, eg,

```
n3->next = 0;
```

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